INCH-POUND
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SUPERSEDING
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PERFORMANCE SPECIFICATION

CLOTH (FOR DRAPERY, BUNK CURTAIN, SLIPCOVER), FIRE RETARDANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers inherently fire retardant, shrink-safe, woven cloth for various shipboard applications such as draperies, bunk curtains, and slipcovers.

1.2 Classification.

1.2.1 Types. Cloth covered by this specification is of the following types, as specified (see 6.2).

a. Type I – Fire retardant organic fibers or blended organic fiber material.

b. Type II – Highly fire retardant inorganic fiber material.

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of publications cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1623 - Fire Performance Requirements and Approved Specifications for Interior Finish Materials and Furnishings (Naval Shipboard Use)

(Copies of this publication are available online at https://quicksearch.dla.mil.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil.

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2.2.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

16 CFR 303 - Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies of this publication are available online at https://www.ecfr.gov/.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9510-AB-ATM-010 - Nuclear Powered Submarine Atmosphere Control Manual; Volume 1

(Copies of the chapter titled "Material Control Program" are available by email request to CommandStandards@navy.mil.)

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this publication are available online via Model Based Product Support (MBPS) at https://mbps.navsea.cloud.navy.mil/Windchill. To request an MBPS account, submit a ticket to the ServiceNow Assistance Portal (SNAP), https://snap.navair.navy.mil/mbps (DoD PKI Certificate required). This publication is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS411-1 - Hazardous Materials Target List

(Copies of this publication are available online at www.aia-aerospace.org.)

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP1 - Gray Scale for Color Change

AATCC EP5 - Fabric Hand: Guidelines for the Subjective Evaluation of

AATCC EP8 - AATCC 9-Step Chromatic Transference Scale

AATCC TM8 - Test Method for Colorfastness to Crocking: Crockmeter

AATCC TM16.3 - Test Method for Colorfastness to Light: Xenon-Arc

AATCC TM61 - Test Method for Colorfastness to Laundering: Accelerated

AATCC TM124 - Test Method for Smoothness Appearance of Fabrics after Home Laundering

AATCC TM135 - Test Method for Dimensional Changes of Fabrics after Home Laundering

(Copies of these publications are available online at https://www.aatcc.org/.)

ASTM INTERNATIONAL

ASTM D578/D578M - Standard Specification for Glass Fiber Strands

ASTM D1424 - Standard Test Method for Tearing Strength of Fabrics by Falling-Pendulum (Elmendorf-Type) Apparatus

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ASTM D3776/D3776M	-	Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D3885	-	Standard Test Method for Abrasion Resistance of Textile Fabrics (Flexing and Abrasion Method)
ASTM D3990	-	Standard Terminology Relating to Fabric Defects
ASTM D5034	-	Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)

(Copies of these publications are available online at https://www.astm.org/.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO/IEC 17025	-	General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of this publication are available online at https://www.iso.org/.)

SAE INTERNATIONAL

SAE AMS-STD-595	-	Colors Used in Government Procurement

(Copies of this publication are available online at (https://www.sae.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Standard sample. When a standard sample is available, as specified (see 6.2), it shall be requested (see 6.3). The finished cloth shall match the standard sample with respect to all characteristics for which the standard sample is referenced.

3.3 Material.

3.3.1 Recycled, recovered, environmentally preferable, or bio-based materials. Recycled, recovered, environmentally preferable, or bio-based materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3.2 Fibers.

3.3.2.1 Type I organic fibers. Organic fibers used in the construction of the cloth shall be permanently fire retardant (PFR) rayon, polyaramid, or any other inherently fire-safe organic fiber or organic fiber blends provided that the cloth woven from the fiber or fiber blends meets all of the requirements of this specification and is consistent with the intended end use application (see 6.1).

3.3.2.2 Type II coated inorganic fibers. Only “C” (Chemical) filament fibrous glass fibers shall be used.

3.3.3 Yarns. Either single or plied yarns shall be used for both type I and type II warp and filling.

3.3.4 Prohibited materials. The cloth shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS411-1 for all other programs. Neither asbestos nor ceramic (refractory) fibers nor material containing any of these fibers shall be used in either type I or type II cloth.

3.4 Weave. Unless otherwise specified (see 6.2), the woven cloth shall be opaque or translucent as required.

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3.5 Color. The color and shade of the finished cloth shall be blue, number 35052 or 35056, in accordance with SAE AMS-STD-595.

3.5.1 Colorfastness. The colorfastness of the cloth to light, wet and dry crocking, heat aging, and laundering shall be as specified in table I.

TABLE I. Colorfastness requirements.

Characteristics	Requirements
Light	The cloth shall show colorfastness to light after 20 standard fading units with a minimum rating of 4 as defined in AATCC EP1 when tested as specified in 4.5.
Crocking (wet and dry)	The cloth shall show colorfastness to wet and dry crocking with a minimum rating of 3.0 as defined in AATCC EP8 when tested as specified in 4.5.
Heat aging	The cloth shall show colorfastness to heat with a minimum rating of 4 as defined in AATCC EP1 when tested as specified in 4.5.1.
Laundering	The cloth shall show colorfastness to laundering with a minimum rating of 3.5 as defined in AATCC EP1 when tested as specified in 4.5.

3.6 Odor. The material shall have no objectionable odor. When tested, the odor concentration shall be 10 or less (see 4.5).

3.7 Finish. The finish shall be type I or II as follows.

3.7.1 Finish (type I). The finish shall approximate that of the standard sample when available. When a standard sample is not available, the finish shall approximate that of the first article sample that has been approved by the contracting officer as conforming to the desired finish.

3.7.2 Finish (type II). A fire retardant, polymeric coating shall be applied to the face and back surfaces of the base cloth to increase the abrasion resistance of the base fabric and shall not exceed 20 percent of the total weight of the coated cloth. The halogen content of the coated cloth shall not exceed 1.75 percent of the total weight of the coated cloth.

3.8 Dimensions and put-up.

3.8.1 Width. The width, inclusive of selvage, shall be as specified (see 6.2), with a tolerance of ±2 percent.

3.8.2 Length. Unless otherwise specified (see 6.2), the cloth shall be in one continuous length of 50±2.5 yards.

3.9 Weave and yarn count. Any weave and yarn count should be used consistent with the total performance requirements of this specification and the intended use of the cloth.

3.10 Physical requirements. When tested, the finished cloth shall conform to the requirements specified in table II (see 4.5).

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TABLE II. Physical requirements.

Characteristics	Requirements	
	Type I	Type II
Weight, ounce/square yard (minimum)	5.2±0.5	7.8±0.5
Breaking strength, pounds (minimum)		
Warp	60	200
Filling	60	175
Abrasion resistance (minimum cycles)	25	25
Tearing strength, pounds (minimum)		
Warp	3.5	18
Filling	3.0	18
Dimensional change (percent change after launderings and cycles when tested [see 4.5])	4.0	2.0

3.11 Fire retardance and optical smoke density. The non-laundered finished cloth shall conform to the fire retardance and optical smoke density requirements specified in MIL-STD-1623 (see 4.5). After five launderings of a finished cloth, the laundered cloth shall also conform to the fire retardance and optical smoke density requirements specified in MIL-STD-1623.

3.12 Hand, character, and appearance. The hand, character, and appearance of the cloth after testing for colorfastness shall show no appreciable change from the original cloth when tested (see 4.5).

3.13 Marking. The length of each roll shall be legibly marked in accordance with 16 CFR 303 with indelible ink on a ticket securely attached to the roll.

3.14 Labels. The cloth shall be labeled in accordance with 16 CFR 303. Labels shall be made of a fire-resistive material conforming to this specification and shall be supplied for later attachment to the finished articles in the quantity specified (see 6.2). Each label shall have the following information:

a. Specification number and type.

b. Manufacturer of the woven cloth.

c. Laundering instructions: Follow manufacturer's recommendations.

d. Lot number.

3.15 Off-gassing. The cloth shall be evaluated for off-gassing in accordance with the requirements of 4.5.2. Based on the circumstances of use and the chemical nature of the cloth, the Navy will determine whether off-gas testing is required or if an administrative assessment is acceptable. In order to be considered acceptable for use in submarines, the cloth shall be assigned to either the "Permitted" or "Limited" category (see 4.5.2 and 6.5).

3.16 Workmanship. The finished cloth shall conform to the requirements of this specification. The demerit points per 100 square yards, when calculated as specified in 4.4, shall not exceed the established maximum point values (see 4.4.1).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

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4.2 First article inspection. First article inspection shall be performed on a finished cloth when a first article sample is required (see 3.1). First article inspection shall consist of the examination and tests specified in 4.2.2, 4.4, and 4.5.

4.2.1 First article sample. If more than one weave, color, or pattern of cloth is acquired at any one time, only one 3-yard cut for each fabric shall constitute the first article sample.

4.2.2 Visual inspection. The first article sample shall be visually examined for weave (see 3.4), color (see 3.5), finish (see 3.7), marking (see 3.13), and labeling (see 3.14).

4.3 Conformance inspection. Conformance inspection shall consist of the examination and tests specified in 4.4 and 4.5. Sampling for inspection shall be performed using the demerit point system (see 4.4.1).

4.4 Examination of an item. Defects found during the examination shall be classified as specified in 4.4.1 through 4.4.5.

4.4.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. Defects defined in accordance with ASTM D3990 that are clearly noticeable at a normal inspection distance of 3 feet shall be scored and assigned demerit points as listed in 4.4.2. The sample size shall be in accordance with table III.

TABLE III. Sample size for yard-by-yard examination.

Lot size (yards)	Sample size (rolls)
3,200 or less	8
3,201 to 10,000	13
10,001 or more	20

(Not more than one roll shall be taken from any shipping container unless the number of shipping containers in the lot is less than the required number of rolls, in which case all shipping containers shall be present in the sample.)

The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceed 50 points. The lot shall be unacceptable if the individual points per 100 square yards of two or more individual rolls exceed 75 points. If one roll exceeds 45 points per 100 square yards, a second sample, the same as the final sample, shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceed 45 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches} \times \text{total yards inspected)}} = \text{Points per 100 square yards}$$

4.4.2 Demerit points. Demerit points shall be assigned as shown in table IV. The following defects, when present in the yard-by-yard and overall examination, shall be scored four points for each yard in which they occur:

a. Baggy, ridgy, or wavy cloth.
b. Width less than specified.
c. Uneven weaving.
d. Odor.
e. Overall uncleanness or discoloration.
f. Character of finish not equal to standard sample or first article when applicable.

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TABLE IV. Demerit points by defect size.

Defect size	Points
Up to 3 inches in any dimension	1
Exceeding 3 inches but not exceeding 6 inches in any dimension	2
Exceeding 6 inches but not exceeding 9 inches in any dimension	3
Exceeding 9 inches in any dimension	4

4.4.3 Examination for length of individual rolls. The individual roll shall be examined for gross length and the number of pieces in the roll. Any gross length found to be more than 2 yards below the gross length marked on the ticket shall be considered a defect with respect to length. Unless otherwise specified (see 6.2), any roll found to contain more than one piece shall be considered a defect with respect to length. The unit of product for this examination shall be one roll.

4.4.4 Examination for total yardage in sample. The lot shall be unacceptable if the total of the actual gross lengths of cloth on rolls in the sample is less than the total of the gross lengths marked on the rolled tickets (see 4.4).

4.4.5 Examination for compliance with 16 CFR 303. During the yard-by-yard examination, each piece in the sample shall be examined for compliance with 16 CFR 303. Each piece not labeled in accordance with 16 CFR 303 shall be a defect. The lot shall be unacceptable if two or more of these defects occur (see 4.4).

4.5 Testing of the end item. The methods of testing specified in table V shall be followed, as applicable. The physical values specified apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test method. The unit of product for test purposes shall be 2 continuous linear yards, full width, of the finished cloth. The sample size shall be as shown in table VI. The lot size shall be expressed in units of 1 linear yard. The lot shall be unacceptable if one or more units of products fail to meet any test requirements specified.

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TABLE V. Examination and tests.

Characteristics	First article	Conformance	Requirement	Test method
Fiber type	X	X	3.3.2	ASTM D578/D578M
Prohibited materials	X		3.3.4	4.5.4
Weave	X		3.4	4.2.2
Color	X		3.5	4.2.2
Colorfastness to:				
Light (170 kilojoules at 420 nanometers or after 20 AFU)	X		Table I	AATCC TM16.3 Option 3 1/, 2/
Heat aging	X		Table I	4.5.1 2/
Laundering (one cycle)	X		Table I	AATCC TM61 test 3A 2/
Crocking (wet and dry)	X		Table I	AATCC TM8 2/
Odor	X	X	3.6	Appendix A
Finish	X		3.7	4.2.2
Abrasion/resistance:				
Warp	X		Table II	ASTM D3885 3/
Filling	X		Table II	ASTM D3885 3/
Dimensional change	X	X	3.8 & Table II	AATCC TM135 (3)(III)(B) (procedure III B)
Weight per square yard	X	X	Table II	ASTM D3776/D3776M
Breaking strength:				
Warp	X		Table II	ASTM D5034
Filling	X		Table II	ASTM D5034
Tearing strength:				
Warp	X		Table II	ASTM D1424
Filling	X		Table II	ASTM D1424
Fire retardance (warp and filling)	X	X	MIL-STD-1623	MIL-STD-1623 4.5.3
Fire retardance after five laundry cycles (warp and filling)	X	X	MIL-STD-1623	AATCC TM135 (3)(III)(B) (procedure III B) and MIL-STD-1623 4.5.3
Hand after colorfastness testing	X		3.12	AATCC EP5
Appearance after colorfastness testing	X		3.12	AATCC TM124 4/
Marking	X	X	3.13	4.4.5
Labeling	X	X	3.14	4.4.5
Off-gassing	X		3.15	4.5.2

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TABLE V. Examination and tests - Continued.

NOTES:

1/ The test shall be terminated if, at the end of exposure of 20 standard fading units, the material shows no appreciable change in color, or shall be compared with the standard sample, if applicable.

2/ Results should be reported as scores defined in 4.4.

3/ End point is the state of destruction.

4/ AATCC three-dimensional smoothness appearance replicas and AATCC TM124 evaluation procedure should be used to grade smoothness appearance after colorfastness testing.

TABLE VI. End item sample size.

Lot size (yards)	Sample size
800 or less	2
801 to 22,000	3
22,001 or more	5

4.5.1 Heat aging test. The test specimen for colorfastness shall be exposed in an oven for 7 days at a temperature of 180±2 °F. The test sample shall be removed, conditioned in accordance with ASTM D3776/D3776M, and evaluated for change in color when compared to a cutting of the initial sample in accordance with AATCC EP1.

4.5.2 Off-gassing. The cloth shall be evaluated for off-gassing in accordance with S9510-AB-ATM-010 chapter titled “Material Control Program” (see 3.15 and 6.5). If the Navy determines that off-gas testing is required, testing shall be conducted at a test facility approved by the Technical Authority (see 3.15 and 6.7). The Navy will review the off-gas test results and assign a usage category. Additionally, the Navy will assign a usage category if an administrative review is conducted in lieu of off-gas testing (see 3.15).

4.5.3 Fire testing provisions. All fire tests specified in this document shall be conducted by an independent testing laboratory that is accredited to ISO/IEC 17025 and is acceptable to the Technical Authority. Accreditation shall be obtained from a recognized accreditation body such as the American Association for Laboratory Accreditation (A2LA) (https://www.a2la.org/) or the International Code Council’s International Accreditation Services (IAS) (https://www.iasonline.org/). The scope of accreditation shall include specific flammability and fire tests. All other fire test provisions shall be as specified (see 6.2 and 6.6).

4.5.4 Prohibited materials. Prohibited materials shall be verified as specified (see 3.3.4 and 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activities within the Military Service or Defense Agency, or within the military service’s system commands. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The woven cloth material detailed in this specification is intended for use as draperies, curtains, and slipcover material on U.S. Navy ships and submarines and for other applications where inherent fire retardance is required. Current stock of woven cloth material may be used until depleted; future requisitions for woven cloth material should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. When first article inspection is required (see 3.1).
d. When a standard sample is available (see 3.2).
e. Requirement for prohibited materials verification (see 3.3.4 and 4.5.4).
f. Whether woven cloth is other than opaque or translucent (see 3.4).
g. Width required (see 3.8.1).
h. Length required, if other than specified (see 3.8.2).
i. Quantity of labels to be supplied (see 3.14).
j. When an individual roll of fabric would contain more than one piece (see 4.4.3).
k. Additional fire testing provisions (see 4.5.3 and 6.6).
l. Packaging requirements (see 5.1).

6.3 Information requests. Information such as patterns, drawings, or standard shade samples of cloth may be accessed by submitting the Defense Supply Center Philadelphia's Specification/Pattern/Drawing Request form online at https://www.dla.mil/Troop-Support/Clothing-and-Textiles/Industry-Support/Spec-Request/. Requests to use equivalent materials or components or to make changes to the pattern should be sent to the contracting officer for approval by the military.

6.4 Safety data sheets (SDS). Contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.5 Material certification. Materials to be installed in submarines are to be controlled to prevent off-gassing, which contaminates the submarine's atmosphere and can result in health hazards to personnel or deleterious effects on machinery. These controls are administered through the Submarine Material Control Program, which is described in the Nuclear Powered Submarine Atmosphere Control Manual, S9510-AB-ATM-010 chapter titled "Material Control Program." Under the Submarine Material Control Program, all materials considered for use on submarines require certification and assignment of a usage category. Under the certification process, candidate materials are selected by Navy activities or contractors, and a request for certification is submitted to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. The certification request is accompanied by detailed information, including descriptions of the material, method of application, usage, and storage. A chemical analysis is conducted, which can be accomplished through off-gas testing. If off-gas testing is required, it must be conducted in a Government approved laboratory. Information pertaining to this test requirement may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. Based on the chemical analysis results, a usage category is assigned to the material defining whether, and to what extent, the material may be used on submarines.

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6.6 Additional fire testing provisions. The Technical Authority reserves the right to witness the tests and perform any of the tests set forth herein where such testing is deemed necessary to assure compliance to prescribed requirements. A2LA (https://www.a2la.org/) or International Code Council's IAS (https://www.iasonline.org/) maintain a full list of the approved laboratories that are accredited to ISO/IEC 17025 and provide details on how to obtain and maintain accreditation (see 6.2).

6.7 Technical authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.8 Subject term (key word) listing.

Designs

Fiber types

Fire performance

Heat aging

Patterns

Woven cloth

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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MEASUREMENT OF ODOR IN ATMOSPHERES (DILUTION METHOD)[1]

A.1 SCOPE

A.1.1 Scope. This Appendix covers a method to determine dilution thresholds of odorous gases discharged from industrial processes and may be used for determining the odor emission rate from stacks and vents. It provides a method for comparing the same source under different operating conditions and for evaluating the efficiency of odor control equipment. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.2 SUMMARY

A.2.1 Summary of test method. A sample of the emission whose odor threshold is to be measured is tested beginning with a dilution beyond the point at which odor can be detected. Increasing concentrations of emission in air are examined until a point is reached at which the odor is consistently detected, each step in the concentration increase consisting of doubling the previous concentration. The ratio of the volume of the diluted sample to the volume of the original sample in the diluted sample is a measure of the detection threshold of the odorous emission sample.

A.3 DEFINITIONS

A.3.1 Odor. The sensation caused by odorants acting on the sense of smell.

A.3.2 Odorant. A substance capable of causing odor sensation when present at a sufficient concentration of air.

A.3.3 Odor detection threshold. The concentration of an odorous emission in the air at which it is just barely sufficient to detect the presence of odorous materials in the air.

A.3.4 Odorous dilution ratio. The concentration of the original odorous emission sample divided by its concentration in the syringe-diluted sample when the latter is at the odor threshold. This ratio, Z, is calculated as:

$$Z=(V_0+V_d)/V_0$$

Where:

V_0 = volume of sample

V_d = volume of the air added to reach odor threshold

(V_0+V_d) = total volume in the syringe

A.3.5 Odorous emission. The emission of an odorant, or multiple odorants capable of producing sensation of odor.

A.3.6 Odorous emission rate. The product of the odorous dilution ratio, Z, and the total volumetric emission rate from the source in cubic meters per minute. It represents the volume of atmosphere that would result every minute if the entire odor emission were diluted to the odor threshold as determined by this test.

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[1] The material in this Appendix was reproduced/adapted from ASTM D1391-78, “Standard Test Method for Measurement of Odor in Atmospheres (Dilution Method),” copyright ASTM International. ASTM D1391-78 was withdrawn by ASTM International in 1986. ASTM International grants the U.S. Naval Sea Systems Command a limited, non-exclusive license to reproduce the content requested from ASTM D1391-78 in the appendix to MIL-PRF-24500B, Cloth (for Drapery, Bunk Curtain, Slipcover), Fire Retardant.

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A.4 PRECAUTIONS

A.4.1 Odor-free test environment. The area used for the test shall be free of interfering odors. An ideal test area is a separate room equipped with activated-carbon filtered air of controlled constant temperature and humidity. All equipment shall be clean and free of odor and be restricted to use for odor determination. An odorless detergent shall be used to clean the equipment. Panel members shall not use perfume or perfumed shaving preparations prior to the test. Panel members shall not smoke, eat, or chew tobacco or gum for at least 1 hour before the test.

A.4.2 Panel member exposure duration. The physical condition of the participants is important. Panel members shall be free of conditions that affect the olfactory system. Prolonged use of the sense of smell causes olfactory fatigue. Repeated smelling of the same odor has the same effect. Therefore, for prolonged testing, rest periods are necessary. Generally, no panel member shall carry out odor testing for longer than 30 minutes to prevent olfactory fatigue. No session shall be longer than 2 hours.

A.4.3 Panel member sensitivity. A total of at least eight panelists of average odor sensitivity shall be chosen. A pre-test can be administered to panelists, with a typical effluent similar to that to be evaluated. It shall be presented to the panel in the same way as described in A.6. If a panelist is found to be significantly insensitive to the odor being studied, they may be excluded from the panel. ("Significantly insensitive" for the purposes of this test means that the panelist cannot detect an odor concentration of four times higher than the most insensitive panel members.)

A.4.4 Test accuracy. At low odorant concentrations, adsorption losses on the glass syringe walls and in the needles reduce the odor. The test should not be used below Z=25, corresponding to 25 odor units per cubic foot. Above Z=50, corresponding to 50 odor units per cubic foot, the test is usually reliable.

A.4.5 Effects due to hydrogen sulfide and high humidity. Some reactive mixtures may undergo catalytic reactions on the syringe walls. Thus, hydrogen sulfide (H_2S) in air and high humidity is catalytically oxidized on ground-glass surfaces. This effect should be considered when applying this method.

A.5 APPARATUS

A.5.1 Apparatus. The following is a list of the apparatus used for the test:

a. Sampling syringe – two or more 100-milliliter Luer-type hypodermic syringes.

b. Transfer syringe – one 2-milliliter and one 10-milliliter Luer-type syringe.

c. Presentation syringe – two 100-milliliter Luer-type syringes for each panel member.

d. Transfer needle – a fitting for connecting the transfer syringe with the sampling and presentation syringes. It is made from two standard 25-gauge BD[3] syringe needles, 1.5 inches long. The mating head of one needle is cut off at a point where its inside bore is equal to the outside diameter of the needle shaft. This mating head is slid over the second needle shaft, with the mating head opening toward the tip of the needle, and silver-soldered in place.

e. Syringe caps – one Luer-type cap for each syringe.

f. Odor-free room maintained at comfortable temperature and humidity.

g. Reagent – odor-free air.

A.6 PROCEDURE

A.6.1 Procedure. The test shall be performed using the following procedures:

a. Fill two sampling syringes with the sample whose odor threshold is to be measured by filling and emptying the syringe twice, inserting the tip into the sample to be tested and pushing the plunger in and out, finally pulling the plunger to the 100-milliliter mark. Place caps on the tips of the syringes and transfer them to the odor-free room for testing.

b. Scrub all syringes and transfer needles with unperfumed detergent. Rinse thoroughly in odor-free tap water, wipe dry with clean cloth or tissue, and allow to dry in the test room atmosphere.

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c. Assign two syringes to each panel member, one for dilution samples and the other for blanks. The blank sample syringes shall never be used for dilution samples. Assign these syringes by use of the manufacturer's code number on the barrel and plunger, or by any other designation. To prevent identification by the panelists, the numbers may be covered with tape.

d. Present odors to the panel in order of increasing concentration, doubling the concentration at each step. The panel leader, or their assistant, prepares the dilution out of sight of the panel. In preparing the dilutions, place the transfer needle on the transfer syringe, remove the cap from the sampling syringe, and with the transfer syringe empty, insert the transfer needle into the sampling syringe and connect the syringes tightly. Withdraw the desired volume of the sample into the transfer syringe. Withdraw the needle from the sampling syringe and recap it. Insert the needle into the presentation syringe, partly filled with air from the odor-free room. Inject the contents of the transfer syringe and then remove the needle from the presentation syringe. Fill the presentation syringe with air from the odor-free room and cap the syringe. If volumes of 2 milliliters or less are required, use the 2-milliliter transfer syringe. If volumes greater than 2 milliliters are required, use the 10-milliliter syringe. Greater dilutions can be made directly with the 2-milliliter syringe by serial dilution. In the same manner, prepare dilution samples for each panel member. Fill each blank syringe with odor-free air from the room. Present the two syringes, dilution and blank, before the panelist to whom they are assigned in a random manner.

e. Each panel member ejects a stream from one syringe in front of their nose and sniffs. Each panelist chooses the manner of ejection by which they feel will produce the most accurate and reproducible results. The panelist ejects gas from the other syringe in a similar manner. The panelist then decides which syringe contains the odor by placing the odorous syringe on a tray or in some other silent manner. The panel leader scores the choice of sample as either right or wrong, and prepares the next set of samples as before, doubling the concentration. When each panelist has chosen correctly at one odor level and two subsequent levels, take the dilution level at which the panelist first made a correct choice as the panelist's odor threshold for that sample.

A.7. REPORT

A.7.1 Report. Following the test, a report shall be made as follows:

a. Calculate the panel detection threshold for the sample as the geometric average of the panelist's dilution level to a logarithm, calculating the arithmetic mean, and determining the anti-logarithm of the mean.

b. Report the panel-averaged detection threshold, number of panelists used, and any unusual observations made.

c. If any panelists are dropped due to lack of sensitivity (A.4), give the number.

d. If any deviations from the details of the method are used, indicate so in the report.

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CONCLUDING MATERIAL

Custodians:
Navy – SH
DLA – CT

Preparing activity:
Navy – SH
(Project 8305-2022-041)

Review activity:
Navy – NU

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST at https://assist.dla.mil.

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